

Problem Sheet 8

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1. Write second order logic formulae to capture the following:
 - (a) There is a path from node s to node t in the graph. The signature is $\tau = \{E, s, t\}$ where s and t are constants.
 - (b) Every bounded non empty set has a least upper bound. The signature is $\tau = \{\leq\}$
2. Let Σ be a finite alphabet. The atomic formulae in MSO defined over Σ^* are $x = y, x < y, S(x, y), X(x)$ and $Q_a(x), a \in \Sigma$. Consider the following logic called MSO_0 having atomic formulae of the following forms:

$$\text{Sing}(X), X \subseteq Y, X < Y, S(X, Y), Q_a(X)$$

where

- $\text{Sing}(X)$ means that X is a SO variable of cardinality 1;
- $X \subseteq Y$ means that every element of the SO variable X is contained in the SO variable Y ;
- $X < Y$ means that SO variables X, Y have cardinality 1, and that the element in Y is greater than the element in X ;
- $S(X, Y)$ means that SO variables X, Y have cardinality 1, and Y contains the successor of the element in X ; and,
- $Q_a(X)$ means that all positions in X are decorated by $a \in \Sigma$.

If φ is an atomic formula in MSO, then $\varphi \wedge \varphi, \neg\varphi, \varphi \vee \varphi, \forall x \varphi$ and $\forall X \varphi$ are formulae in MSO. Similarly, if φ is an atomic formula in MSO_0 , then, $\varphi \wedge \varphi, \neg\varphi, \varphi \vee \varphi$ and $\forall X \varphi$ are formulae in MSO_0 .

Compare the expressiveness of MSO and MSO_0 .

3. For the formula $\exists x \forall y (x < y \rightarrow Q_a(y))$ give an equivalent MSO_0 formula. Also draw the equivalent DFA following the steps done in class.
4. Consider the following NFA $N = (\{0, 1, 2, 3\}, \{a, b\}, \Delta, \{0\}, \{1\})$ with $\Delta(0, b) = \{1\}, \Delta(1, a) = \{2\}, \Delta(2, a) = \{2\}, \Delta(2, b) = \{3\}$ and $\Delta(3, b) = \{0\}$. Write an MSO formula with two SO variables that characterizes $L(N)$.
5. **Theorem 1 (Propositional Compactness Theorem)** *Let S be a set of propositional formulae. S is satisfiable iff every finite subset of S is satisfiable.*

Prove the above theorem using soundness and completeness of natural deduction.

6. **Theorem 2 (Compactness theorem for FOL)** *Let S be a set of formulae in first-order logic. S is satisfiable iff every finite subset of S is satisfiable.*

Using the fact that FOL is semi-decidable, prove the compactness theorem for FOL.

7. Earlier, we saw how to write the following in MSO: “There is a path from node s to node t in the graph” using the signature $\tau = \{E\}$.

Show, using compactness theorem, that you cannot capture this using FOL with the same signature.